

Catastrophic General Health Out-of-Pocket Expenditure in Nepal: No-Addback OOPG Analysis

Authors' calculations from Nepal Living Standards Survey IV

Abstract

This draft examines catastrophic out-of-pocket spending for general health events in Nepal using NLSS IV. The analysis focuses on OOPG, defined as communicable disease or injury spending reported in Section 8(B). We present three complementary denominators: total household expenditure, nonfood expenditure, and adult-equivalence-adjusted capacity to pay. In this rerun, OOPG is not added back to the consumption denominators used for CHE or CTP.

1 Introduction

Financial protection is a central objective of health systems because health payments can force households to sacrifice other consumption, borrow, sell assets, or fall into poverty. Catastrophic health expenditure (CHE) measures this burden by comparing out-of-pocket payments with household resources. This paper focuses on OOPG, a measure of general health out-of-pocket spending related to communicable disease or injury. This scope is useful because it is directly observed over a recent 30-day recall period in NLSS IV and allows a focused assessment of acute health-payment burden.

The analysis follows the catastrophic-payment tradition in O'Donnell et al. (2008) and Wagstaff and van Doorslaer (2003), while also drawing on the WHO/Xu capacity-to-pay approach (Xu et al., 2003; Xu, 2005). The total-expenditure measure is retained because it is transparent and comparable with the SDG 3.8.2 convention, which monitors health spending above 10% or 25% of household consumption or income (World Health Organization, 2026). The capacity-to-pay extension is added because total expenditure may not represent discretionary resources equally across households with different subsistence needs.

2 Data and OOPG Measure

The data are from Nepal Living Standards Survey IV. OOPG is constructed by summing individual Section 8(B) payments for communicable disease or injury to the household level. The household is the analytical unit because catastrophic expenditure is a budget event: the payment may originate with one member, but the financial burden is absorbed by the household.

NLSS welfare aggregates exclude health spending by construction. For this no-addback rerun, nominal household consumption is reconstructed from official food and nonfood welfare components and used directly as the total-expenditure CHE denominator. This follows the requested specification in which OOPG remains only in the numerator.

3 Methods

3.1 Total-expenditure CHE

Let T_i denote household OOPG and x_i denote reconstructed monthly nominal household consumption without OOPG addback. The total-expenditure share is

$$s_i^T = \frac{T_i}{x_i}, \quad x_i = \text{nominal consumption}_i. \quad (1)$$

A household is catastrophic at threshold z if $s_i^T > z$. We report multiple thresholds, with the 10% threshold emphasized for interpretation and comparability.

3.2 Nonfood CHE

The second specification uses nonfood expenditure as a practical ability-to-pay proxy, following the Chapter 18 framework in O'Donnell et al. (2008). Let c_i^{NF} denote real monthly nonfood consumption without OOPG addback. The nonfood share is

$$s_i^{NF} = \frac{T_i^r}{c_i^{NF}}, \quad (2)$$

where T_i^r is real monthly OOPG. This denominator is narrower than total expenditure and therefore gives a stricter view of financial stress.

3.3 Adult-equivalence capacity to pay

The third specification follows the logic of the WHO/Xu capacity-to-pay method (Xu et al., 2003; Xu, 2005). Adult equivalence is used only here because it is a way of estimating household subsistence needs, not a way of rescaling the health bill. If both OOPG and total consumption were divided by the same equivalence scale, the scale would cancel: $(T_i/E_i)/(x_i/E_i) = T_i/x_i$. Comparing household OOPG with consumption per adult equivalent would mix units by placing a household-level payment over a one-adult-equivalent denominator.

Adult equivalence therefore enters through subsistence expenditure. We use the project scale

$$E_i = (A_i + 0.5K_i)^{0.75}, \quad (3)$$

where A_i is the number of adults aged 15 years or older and K_i is the number of children below age 15. Food expenditure per adult equivalent is computed as

$$f_i^e = \frac{f_i}{E_i}. \quad (4)$$

The subsistence line is the household-weighted mean of f_i^e among households in the 45th–55th percentiles of the no-addback food-share distribution. In this analysis, that line is NPR 7,894 per adult equivalent per month. Household subsistence expenditure is this line multiplied by E_i . Capacity to pay is then

$$C_i = \begin{cases} x_i - \text{subs}_i, & \text{if } f_i \geq \text{subs}_i, \\ x_i - f_i, & \text{if } f_i < \text{subs}_i. \end{cases} \quad (5)$$

The CTP catastrophic share is T_i/C_i . This approach follows the role of equivalence scales described by Koch et al. (2018): equivalence affects the poverty line, subsistence expenditure, and capacity to pay, not the OOP numerator itself.

3.4 Equity and impoverishment measures

For each denominator, we report headcount, overshoot, and mean positive overshoot. Distribution-sensitive measures use concentration indices and rank-weighted headcounts/overshoots. Because this no-addback rerun treats OOPG as already included in observed consumption, households are ranked by pre-OOPG welfare, defined as monthly official real per-capita consumption, $pcep/12$. Impoverishment is calculated by comparing this pre-OOPG welfare with post-OOPG welfare after subtracting monthly real per-capita OOPG, defined as household monthly OOPG divided by household size and deflated by the Paasche index. Post-OOPG welfare is floored at zero; 42 households required this floor because raw subtraction was negative.

4 Results

Table 1: Incidence and intensity of catastrophic OOPG payments

Measure	5%	10%	15%	25%	40%
<i>OOPG as a share of total expenditure</i>					
Headcount	15.83%	8.39%	5.12%	2.66%	1.47%
Standard error	0.69%	0.44%	0.32%	0.22%	0.15%
Overshoot	2.45%	1.88%	1.56%	1.19%	0.91%
Mean positive overshoot	15.48%	22.42%	30.39%	44.90%	61.51%
<i>OOPG as a share of nonfood expenditure</i>					
Headcount	—	—	11.45%	6.77%	4.12%
Standard error	—	—	0.57%	0.39%	0.28%
Overshoot	—	—	5.34%	4.46%	3.67%
Mean positive overshoot	—	—	46.61%	65.82%	89.17%

Notes: Estimates use household survey weights. OOPG is Section 8(B) communicable disease or injury out-of-pocket spending. The total-expenditure denominator is reconstructed nominal monthly consumption without adding OOPG back.

At the 10% total-expenditure threshold, 8.39% of households incurred catastrophic OOPG spending. The associated mean positive overshoot was 22.42%, indicating that households crossing the threshold were not merely marginally above it. Under the 40% nonfood threshold, the headcount was 4.12%, reflecting a stricter ability-to-pay interpretation.

Table 2 applies the adult-equivalence capacity-to-pay denominator. At the 10% threshold, 16.52% of households crossed the CTP line; this should be read as a lower-threshold financial-stress indicator, because even one-tenth of capacity to pay can be consequential once subsistence needs are protected. The 25% and 40% thresholds are stricter: 6.88% and 4.06% of households crossed them, respectively. These estimates should be read alongside, not instead of, the total-expenditure estimate. They ask a different welfare question: how large is OOPG after protecting basic subsistence needs?

Table 2: Adult-equivalence capacity-to-pay catastrophic OOPG payments

Measure	10% CTP	25% CTP	40% CTP
Headcount	16.52%	6.88%	4.06%
Standard error	0.77%	0.41%	0.29%
Overshoot	6.06%	4.47%	3.68%
Mean positive overshoot	36.71%	64.94%	90.53%

Notes: Capacity to pay is reconstructed nominal consumption without OOPG addback less adult-equivalence-adjusted subsistence expenditure, except when observed food expenditure is below subsistence, in which case observed food expenditure is subtracted. Estimates use household survey weights.

OOPG Total and Nonfood Budget Share against Cumulative Percentage of Households Ranked by Decreasing Budget Share, Nepal NLSS IV 2022/23

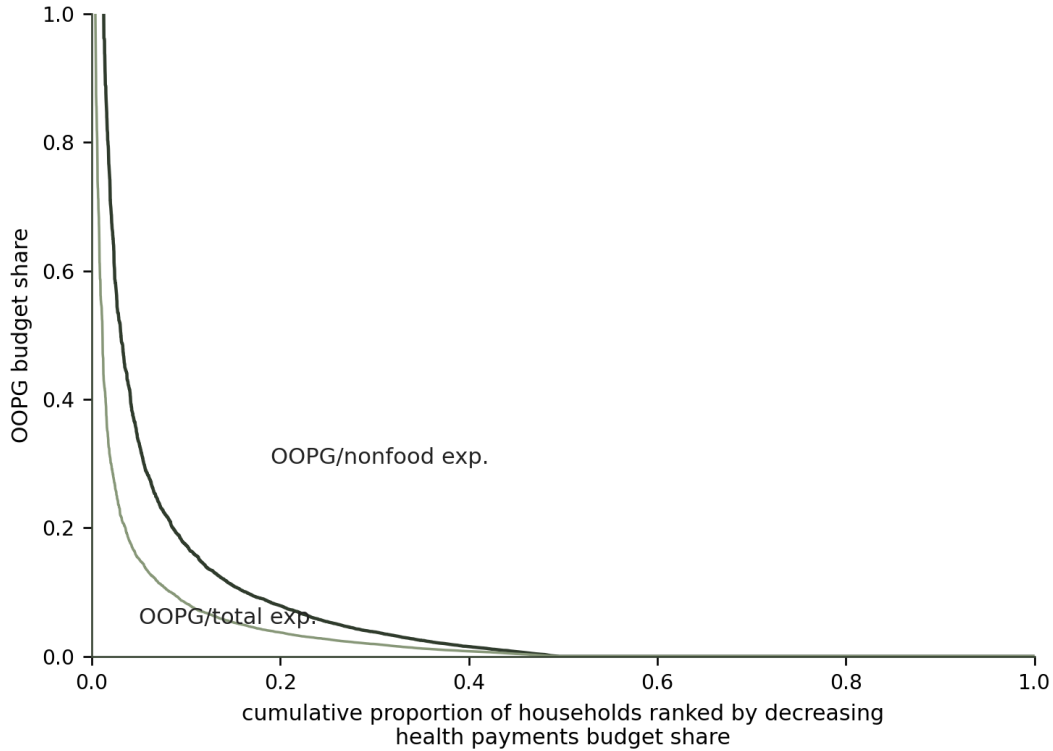


Figure 1: OOPG budget-share curve

Figure 1 shows the distribution of OOPG budget shares. The left tail identifies households for which OOPG absorbs a particularly large share of resources, while the flatter portion of the curve shows that most households have relatively low OOPG shares.

Table 3: Distribution-sensitive OOPG catastrophic-payment measures

Measure	5%	10%	15%	25%	40%
<i>OOPG as a share of total expenditure</i>					
Concentration index, C^H	-0.1606	-0.1850	-0.2005	-0.2400	-0.2221
Rank-weighted headcount, H^W	18.37%	9.95%	6.15%	3.30%	1.80%
Concentration index, C^O	-0.1936	-0.1996	-0.2014	-0.1952	-0.1843
Rank-weighted overshoot, O^W	2.92%	2.26%	1.87%	1.43%	1.07%
<i>OOPG as a share of nonfood expenditure</i>					
Concentration index, C^H	—	—	-0.2118	-0.2268	-0.2343
Rank-weighted headcount, H^W	—	—	13.88%	8.31%	5.08%
Concentration index, C^O	—	—	-0.2419	-0.2459	-0.2489
Rank-weighted overshoot, O^W	—	—	6.63%	5.55%	4.59%
<i>OOPG as a share of adult-equivalence capacity to pay</i>					
Concentration index, C^H	—	-0.2530	—	-0.3223	-0.3364
Rank-weighted headcount, H^W	—	20.70%	—	9.09%	5.43%
Concentration index, C^O	—	-0.3298	—	-0.3453	-0.3476
Rank-weighted overshoot, O^W	—	8.06%	—	6.01%	4.95%

Notes: Distribution-sensitive measures use household weights and rank households by pre-OOPG welfare, defined as monthly official real per-capita consumption under the no-addback assumption.

Table 3 shows how the interpretation changes when incidence is read alongside welfare rank. Positive concentration indices indicate that the relevant catastrophic measure is more concentrated among better-off households, while negative values indicate concentration among poorer households. For the 10% total-expenditure headcount, the concentration index is -0.1850, and the rank-weighted headcount is 9.95%. For the 40% nonfood headcount, the concentration index is -0.2343, with a rank-weighted headcount of 5.08%. Under the CTP approach, the 40% CTP concentration index is -0.3364. This pattern should be interpreted carefully: lower observed catastrophic spending among poorer households may reflect lower service use or lower ability to pay, not necessarily protection from need.

Table 4 reports the no-addback poverty-impact calculation. Pre-OOPG welfare is monthly official NLSS *pcep*/12, while post-OOPG welfare subtracts monthly real per-capita OOPG from *pcep*/12 with a zero floor. The poverty line is also converted to monthly terms as *pline*/12. The table indicates that OOPG raises the poverty headcount from 20.27% to 23.40%, a change of +3.13 pp, corresponding to approximately 898,800 people pushed below the poverty line.

5 Discussion

The three denominators answer related but distinct questions. The total-expenditure measure provides an intuitive budget-share headline. The nonfood measure narrows the denominator to resources outside food consumption and therefore moves closer to an ability-to-pay interpretation. The adult-equivalence CTP measure is the most welfare-sensitive because it explicitly protects a

Table 4: OOPG-associated poverty impact

Measure	Pre-OOPG	Post-payment	Change
Poverty headcount	20.27%	23.40%	+3.13 pp
Poverty gap (monthly NPR)	275	354	+80
Normalized poverty gap	4.52%	5.83%	+1.31 pp
Mean positive gap (monthly NPR)	1,355	1,514	+160
Mean positive normalized gap	22.30%	24.92%	+2.63 pp
People pushed below poverty line	–	–	898,800
Households crossing poverty line	–	–	264

Notes: Estimates use individual survey weights. Pre-OOPG welfare is monthly official NLSS pcep (*pcep/12*) under the no-addback assumption. Post-OOPG welfare subtracts monthly real per-capita OOPG from *pcep/12* and is floored at zero where raw subtraction would be negative.

household-specific subsistence requirement before evaluating health-payment burden.

The adult-equivalence adjustment is not used in the simple total-expenditure ratio because it would either cancel algebraically or create an inconsistent comparison of household payments with per-equivalent-adult consumption. Its appropriate role is in the construction of capacity to pay, where household composition affects the estimated resources needed for basic living.

References

- Koch, K. J., Pedraza, C. C., and Schmid, A. (2018). Out-of-pocket expenditure and financial protection in the health sector: A systematic review of equivalence scale sensitivity. *Health Policy and Planning*, 33(8):966–977.
- O’Donnell, O., van Doorslaer, E., Wagstaff, A., and Lindelow, M. (2008). *Analyzing Health Equity Using Household Survey Data: A Guide to Techniques and Their Implementation*. World Bank, Washington, DC.
- Wagstaff, A. and van Doorslaer, E. (2003). Catastrophe and impoverishment in paying for health care: With applications to vietnam 1993–1998. *Health Economics*, 12(11):921–934.
- World Health Organization (2026). Population with household expenditures on health greater than 10 percent of total household expenditure or income (sdg 3.8.2). [https://www.who.int/data/gho/data/indicators/indicator-details/GHO/total-population-with-household-expenditures-on-health-greater-than-10-of-total-household-expenditure-or-income-\(sdg-3-8-2\)-\(-\)](https://www.who.int/data/gho/data/indicators/indicator-details/GHO/total-population-with-household-expenditures-on-health-greater-than-10-of-total-household-expenditure-or-income-(sdg-3-8-2)-(-)). Accessed 28 May 2026.
- Xu, K. (2005). Distribution of health payments and catastrophic expenditures: Methodology. Technical Report EIP/FER/DP.05.2, World Health Organization, Geneva.
- Xu, K., Evans, D. B., Kawabata, K., Zeramardini, R., Klavus, J., and Murray, C. J. L. (2003). Household catastrophic health expenditure: A multicountry analysis. *The Lancet*, 362(9378):111–117.